

1. Executive Summary

An investigation into workstation **WS01-ENG** assigned to employee John Doe revealed unauthorized staging and exfiltration of NorthBridge Manufacturing proprietary CAD designs (**turboblade_v3_draft.step**)[cite: 2]. The incident occurred between July 27 and July 28, 2026, culminating in a 14.2 MB encrypted transfer to an external C2/dropzone host (**198.51.100.45**)[cite: 2]. Chain of custody has been maintained without integrity deviations[cite: 2].

2. Evidence Summary & Chain of Custody

All analyzed evidence was validated using SHA-256 cryptographic hashing prior to and upon completion of forensic analysis[cite: 1, 2]:

- **ws01_workstation_img.e01** (EVD-DISK-001): **e3b0c44298fc...**[cite: 2]
- **ws01_memory_dump.raw** (EVD-MEM-001): **8743b52063cd...**[cite: 2]
- **perimeter_fw_capture.pcap** (EVD-NET-001): **4f53cda18c2b...**[cite: 2]
- **jdoe_mailbox_backup.pst** (EVD-EML-001): **1a2b3c4d5e6f...**[cite: 2]

3. Detailed Forensic Findings

A. Initial Contact (Email Domain)

On 2026-07-27 at 08:30:12 UTC, **jdoe@northbridge.local** received an inbound message with Subject *"Career Opportunity - Senior CAD Architect"* (**EVD-EML-001**)[cite: 2]. Header inspection revealed failed SPF validation, establishing communication origin from an unauthorized recruitment entity[cite: 2].

B. Access & Data Staging (Disk & Memory Domains)

- **USB Storage Connection:** On 2026-07-28 at 14:02:11 UTC, a Kingston DataTraveler USB (Serial **8A23-44DF**) was attached (**EVD-DISK-001**)[cite: 2].

- **File Access:** LNK artifacts confirm `C:\\Projects\\CAD\\turboblade_v3_draft.step` was accessed at 14:15:22 UTC (EVD-DISK-001)[cite: 2].
- **Process Execution:** Volatility 3 memory inspection confirmed `explorer.exe` spawned `cmd.exe` (PID 4812), which subsequently launched `7z.exe` (PID 5120) with command line `7z.exe a -pSecureCAD! archive.7z C:\\Projects\\CAD\\*.step` (EVD-MEM-001)[cite: 2].

C. Network Exfiltration (Network Domain)

Packet capture analysis reveals TCP stream #14 initiated by PID 6012 (`curl.exe`) at 14:35:10 UTC to destination `198.51.100.45:443` transmitting 14.2 MB[cite: 2]. This matches the staging parameters and size of `archive.7z` (EVD-NET-001)[cite: 2].

D. Anti-Forensic Deletion (Disk Domain)

At 14:45:00 UTC, \$MFT record #42109 recorded deletion status for `turboblade_v3_draft.step` and local archive files (EVD-FILE-001)[cite: 2]. Carving techniques successfully recovered the intact file[cite: 2].

4. Synthesis & Attack Narrative

The incident progressed through four discrete, chronologically verifiable stages:

1. **Solicitation:** External email contact established baseline communication (EVD-EML-001)[cite: 2].
2. **Access:** Unauthorized hardware attachment and target CAD file staging (EVD-DISK-001)[cite: 2].
3. **Compression:** Password-protected archiving in volatile memory space (EVD-MEM-001)[cite: 2].
4. **Exfiltration & Deletion:** Outbound HTTPS transfer followed by local file deletion (EVD-NET-001, EVD-FILE-001)[cite: 2].

5. Limitations & Threat to Validity

Network payloads were TLS-encrypted; exfiltrated payload contents were correlated via temporal match and exact archive file sizes rather than plaintext payload reconstruction[cite: 1, 2].

6. Recommendations

1. Revoke workstation access credentials for account `jdoe`.

2. Implement Endpoint USB storage whitelisting via Group Policy.
3. Restrict endpoint command-line network execution utilities (`curl.exe`, `powershell.exe`). """,

"06_Defense_Deck/Courtroom_Walkthrough_Presentation.md": """"# MOCK COURTROOM
DEFENSE SLIDES **Case:** NorthBridge Manufacturing v. Incident Subject
Project: CAP-DFE-3W Capstone Defense
Standard: Court-Admissible Forensic Analysis (NIST SP 800-86)
